

Experimental Error – Problem Set

Complete the homework in your science notebook. Make sure to **show all work**. Make sure your final error calculations have the appropriate units.

1. When you do an experiment multiple times, what do statistics tell you about the data you collect?
2. How does doing an experiment multiple times increase the reliability of your data?
3. Explain what a trendline is and why you use it on a scatterplot.
4. Name the 5 sources of error:
5. When analyzing data, you want precision and accuracy. What are they?
6. What is an “outlier” and what should you do if you get one in your experiment?
7. Why is percent error more useful than just error?

Use these equations to solve the following 2 problems:

$$\% \text{ Error} = |\text{measured} - \text{actual}| \div \text{actual} \times 100\%$$

8. You estimate the height of your teacher to be 160 cm, but it is actually 165 cm. What is your error? What is your % error?

Identify what variables you have	Identify what variables you are looking for	Identify the equation you will use	Plug your variables in and solve	Answer with units
Height _{actual} = Height _{measured} =	Error =			

Identify what variables you have	Identify what variables you are looking for	Identify the equation you will use	Plug your variables in and solve	Answer with units
Height _{actual} = error =	% Error =			

9. A student measured the temperature of boiling water and got an experimental reading of 97.5°C . The actual temperature of water's boiling point is 100°C . What is the error? What is the percent error?

Identify what variables you have	Identify what variables you are looking for	Identify the equation you will use	Plug your variables in and solve	Answer with units
$\text{Temp}_{\text{actual}} =$ $\text{Temp}_{\text{measured}} =$	Error =			

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$\text{Temp}_{\text{actual}} =$ error =	% Error =			